



UNIVERSITY OF THE FREE STATE

BLOEMFONTEIN CAMPUS

CSIS3744

**DEPARTMENT OF COMPUTER SCIENCE AND
INFORMATICS**

CONTACT NUMBER: 051 401 3702

**EXAMINATION: Main End-of-Year 2021
PAPER 1**

ASSESSOR: 1. Prof Okuthe P. Kogeda

MODERATOR: 1. Mnr. W.S.J. Marais

TIME: 3 HOURS

MARKS: 100

INSTRUCTIONS: Answer ALL the questions. Write legibly. Number your questions clearly and correctly.

Question 1 [16 Marks]

Discuss the difference between the following terms and concepts:

- (a) Static routing and dynamic routing. [2 Marks]
- (b) Recursive and iterative DNS query. [2 Marks]
- (c) Dynamic and Static ARP table entries. [2 Marks]
- (d) Persistent agent and non-persistent agent. [2 Marks]
- (e) Data Terminal Equipment (DTE) and Data Circuit terminating Equipment (DCE). [2 Marks]
- (f) Core and Edge routers. [2 Marks]
- (g) Bandwidth and Throughput. [2 Marks]
- (h) Unidirectional antenna and omnidirectional antenna. [2 Marks]

Question 2 [26 Marks]

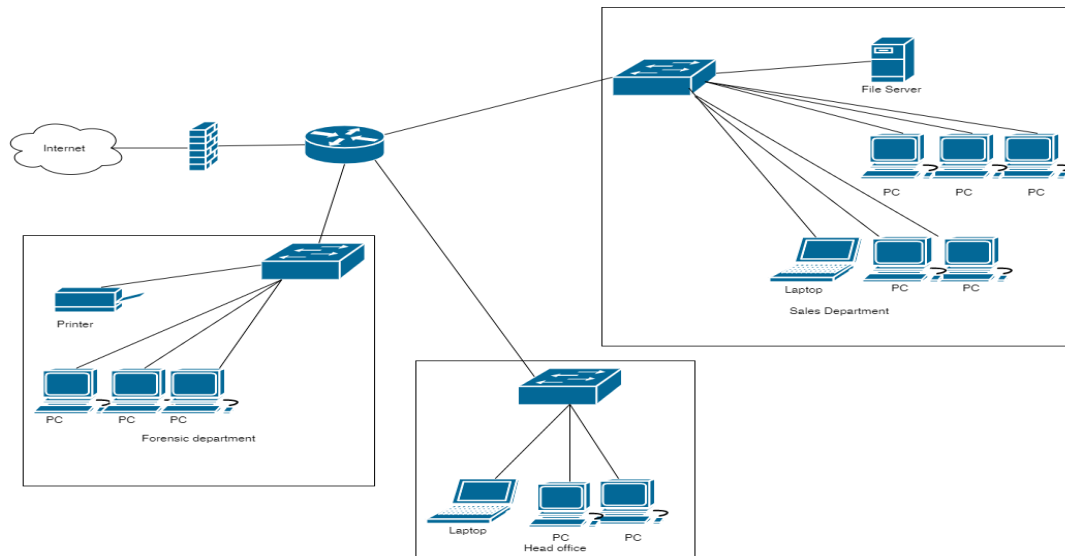
The Defy Café makes use of the public network address, 192.169.4.0

- (a) What network class does the above-mentioned network address belong to? [1 Mark]
- (b) What are five different reasons for using VLANs? [5 Marks]
- (c) Provide and explain THREE common configuration errors that can occur when configuring VLANs. [6 Marks]
- (d) An attacker configures a VLAN frame with two tags instead of just one. The first tag directs the frame to the authorized VLAN. After the frame enters the first VLAN, the switch appropriately removes the tag, then discovers the next tag, and sends the frame along to a protected VLAN, which the attacker is not authorized to access. What kind of attack is this? [2 Marks]
- (e) What kinds of issues might indicate a misconfigured ACL? [3 Marks]
- (f) The Defy café wants to divide their network into two subnets.
 - i. Calculate the new subnet mask for the proposed sub networks. [3 Marks]
 - ii. How many usable addresses are available per subnet? [2 Marks]
 - iii. Complete the table below by supplying each subnet's network address, broadcast address and usable host address range: [4 Marks]

	Subnet	Network address	Broadcast address	Usable host address range
(i)	1			
(ii)	2			

Question 3 [26 Marks]

mojaPay digital payment services has forensic department, sales department and a head office. See an illustration of this network in the diagram below.



- (a) mojaPay digital payment services realized some security threats, although they have a firewall. Identify two security threats that this network could suffer. [2 Marks]
- (b) Discuss the mitigation procedures for the identified threats in Question (a). [2 Marks]
- (c) As a network administrator of this mojaPay digital payment services, you want to determine which computer is the host. What command-line utility might you use to do this? [2 Marks]
- (d) Discuss the IS-IS link state routing protocol. [4 Marks]
- (e) A PC at the sales department is unable to access the network. When you check the LED lights near the PC's network port, you discover the lights are not lit.
 - i. Which layer of the OSI 7 layer reference model are you using to troubleshoot this problem? [1 Mark]
 - ii. At which two layers does the network interface card work? [2 Marks]
 - iii. Why is it important to use a structured cabling standard when installing and managing cabling systems? [2 Marks]
- (f) Explain in detail the role that the DNS protocol played during the transmission of the request with regard to local, top level and root name servers. [4 Marks]
- (g) Some of the employees of CHADEMA financial services connect to a LAN from a remote location using the Point-to-Point Protocol (PPP).
 - i. What server is necessary for this connection? [1 Mark]
 - ii. Which secured tunnelling protocol might be able to cross firewalls where IPsec is blocked? [1 Mark]

- iii. Why is Point-to-Point Protocol (PPP) a better protocol than Serial Line Internet Protocol (SLIP)? [5 Marks]

Question 4 [22 Marks]

St Andrews school want to ensure that all the data about its operations are available to academic staff, support staff, learners and visitors to their site. They have therefore decided to solicit for cloud hosting services. You have been contracted to help with the implementation of such a system.

- (a) List and explain some of the cloud deployment models that you would discuss with St Andrews management? [8 Marks]
- (b) As you discuss with St Andrews management team, you receive a lot of questions, which prompted you to explain to them the features that all cloud services usually have in common. Explain FOUR of the features you discussed. [4 Marks]
- (c) St Andrews school system would involve data at rest, in use and in motion. They want to use encryption methods to enforce security. However, they do not know the three benchmarks you can use to evaluate any encryption method. Discuss what would be your response? [6 Marks]
- (d) St Andrews school want to involve streaming of audio and video as part of providing lessons to learners. However, they want a secure protocol, which also would provide better performance. In this regard, you advised them to use datagram transport layer security protocol. Explain why? [2 Marks]
- (e) Explain two reasons why wireless networks experience a greater reduction in throughput as compared to wired networks. [2 Marks]

Question 5 [10 Marks]

UDP and TCP are the main protocols used on the transport layer of the OSI model.

- (a) You decided to use TCP for your implementation; however, later you realized performance degradation of the network. You decided to change to UDP. Explain how this change would improve the performance of the network? [3 Marks]
- (b) Explain why DTLS relies on UDP instead of TCP to perform security functions. [2 Mark]
- (c) IPv6 has two different types of unicast addresses. How do these two types differ from each other? [4 Marks]
- (d) What is MAC address spoofing? [1 Mark]

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